

5ME42A(ME) - Manufacturing Technology

For 5th Semester B.E. Mechanical Engineering

Dr. Kailash Chaudhary

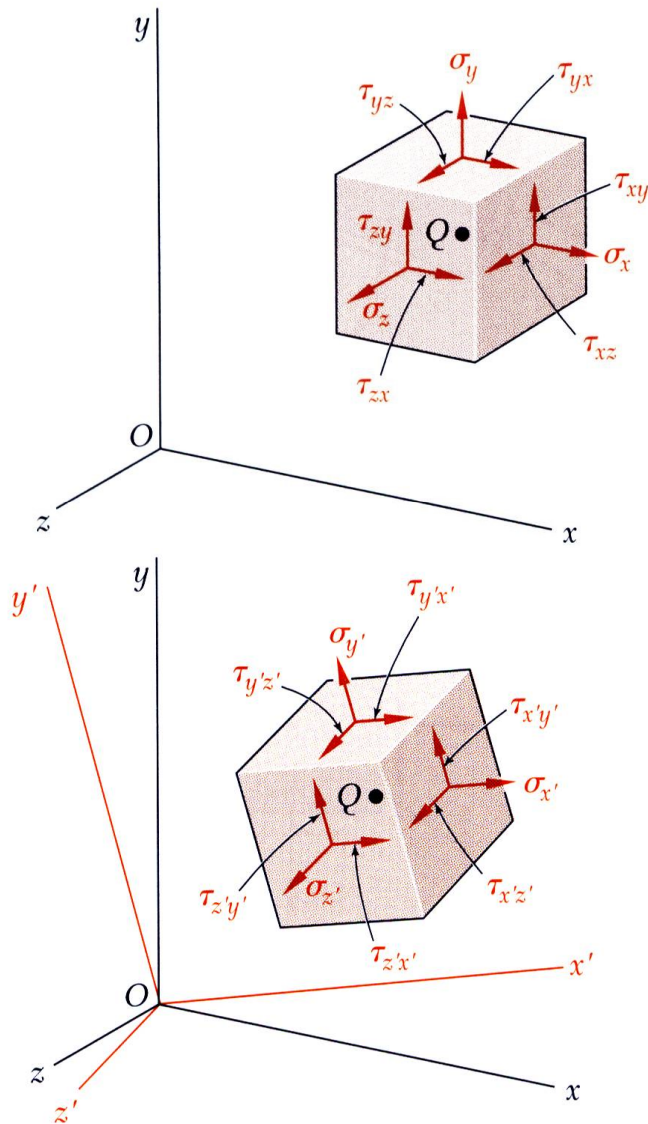
Ph.D. (Mechanical Design), M.E. (P & I), B.E. (Mechanical Engg)

Assistant Professor

Department of Mechanical Engineering

MBM University Jodhpur

Principal Stress and Strain



- The most general state of stress at a point may be represented by 6 components

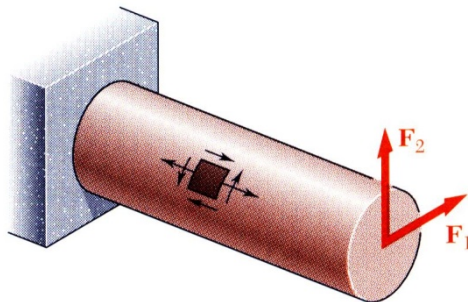
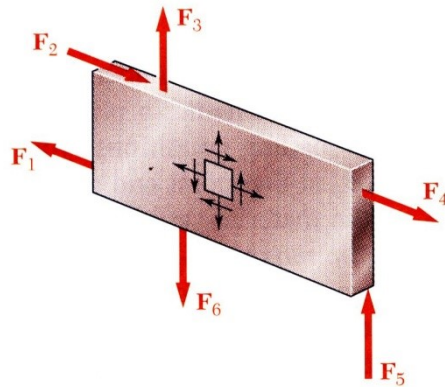
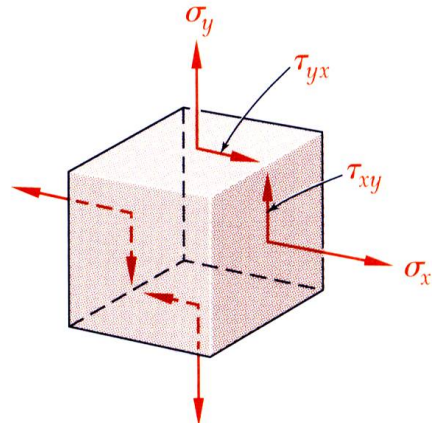
$\sigma_x, \sigma_y, \sigma_z$ normal stresses

$\tau_{xy}, \tau_{yz}, \tau_{zx}$ shearing stresses

(Note : $\tau_{xy} = \tau_{yx}$, $\tau_{yz} = \tau_{zy}$, $\tau_{zx} = \tau_{xz}$)

- Same state of stress is represented by a different set of components if axes are rotated.

Known as ***Transformation of Stresses and Strains***

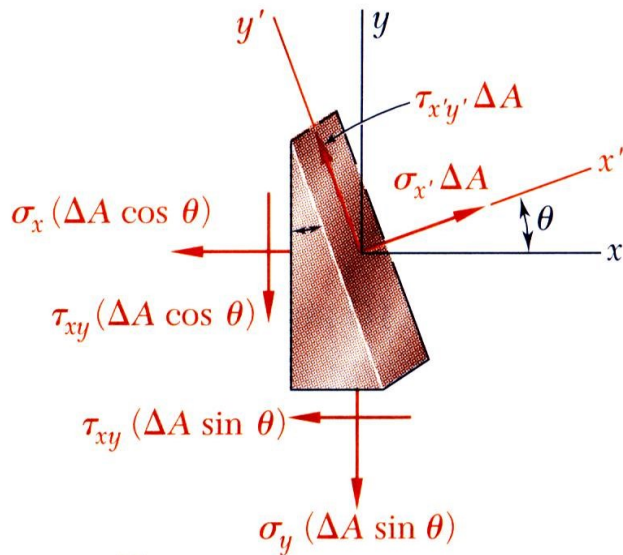


- **Plane Stress** - state of stress in which two faces of the cubic element are free of stress. For the illustrated example, the state of stress is defined by

$$\sigma_x, \sigma_y, \tau_{xy} \quad \text{and} \quad \sigma_z = \tau_{zx} = \tau_{zy} = 0.$$

- State of plane stress occurs in a thin plate subjected to forces acting in the midplane of the plate.
- State of plane stress also occurs on the free surface of a structural element or machine component, i.e., at any point of the surface not subjected to an external force.

Transformation of Plane Stress



- Consider the conditions for equilibrium of a prismatic element with faces perpendicular to the x , y , and x' axes.

$$\begin{aligned} \sum F_{x'} = 0 = & \sigma_{x'} \Delta A - \sigma_x (\Delta A \cos \theta) \cos \theta - \tau_{xy} (\Delta A \cos \theta) \sin \theta \\ & - \sigma_y (\Delta A \sin \theta) \sin \theta - \tau_{xy} (\Delta A \sin \theta) \cos \theta \end{aligned} \quad (1)$$

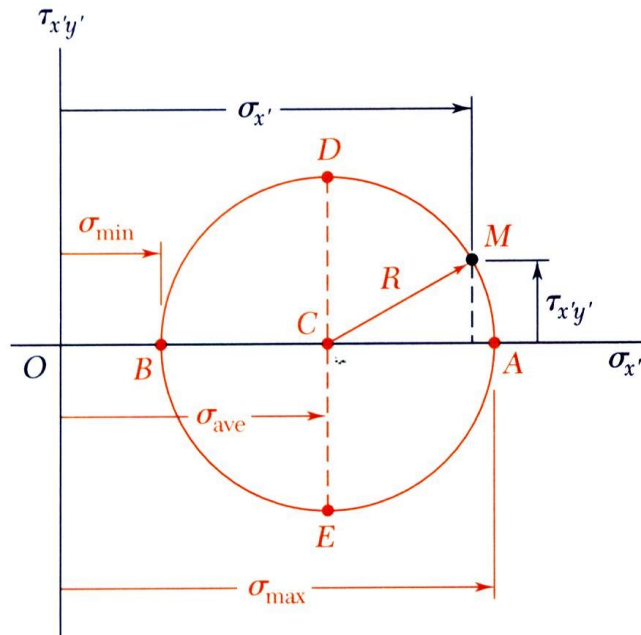
$$\begin{aligned} \sum F_{y'} = 0 = & \tau_{x'y'} \Delta A + \sigma_x (\Delta A \cos \theta) \sin \theta - \tau_{xy} (\Delta A \cos \theta) \cos \theta \\ & - \sigma_y (\Delta A \sin \theta) \cos \theta + \tau_{xy} (\Delta A \sin \theta) \sin \theta \end{aligned} \quad (2)$$

- The equations may be rewritten to yield

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta \quad (3)$$

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta \quad (4)$$

Principal Stresses



- The previous equations are combined to yield parametric equations for a circle,

$$(\sigma_{x'} - \sigma_{ave})^2 + \tau_{x'y'}^2 = R^2 \quad (5)$$

where

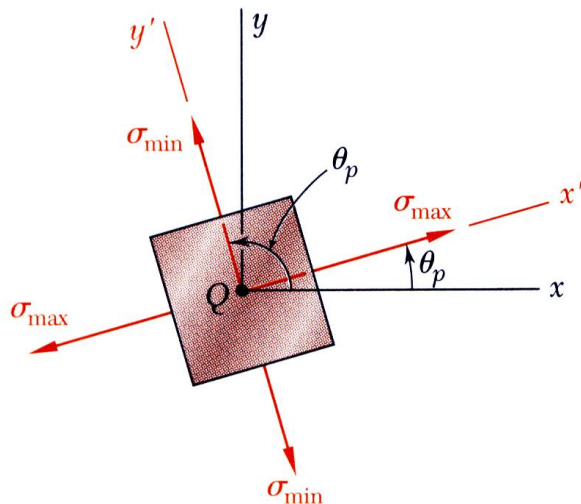
$$\sigma_{ave} = \frac{\sigma_x + \sigma_y}{2} \quad R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

- Principal stresses** occur on the **principal planes of stress** with zero shearing stresses.

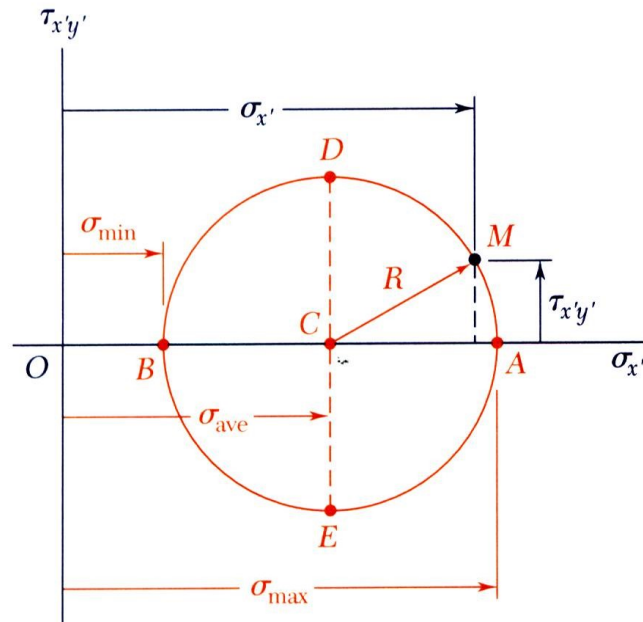
$$\sigma_{max,min} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \quad (6)$$

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} \quad (7)$$

Note : defines two angles separated by 90°



Maximum Shearing Stress



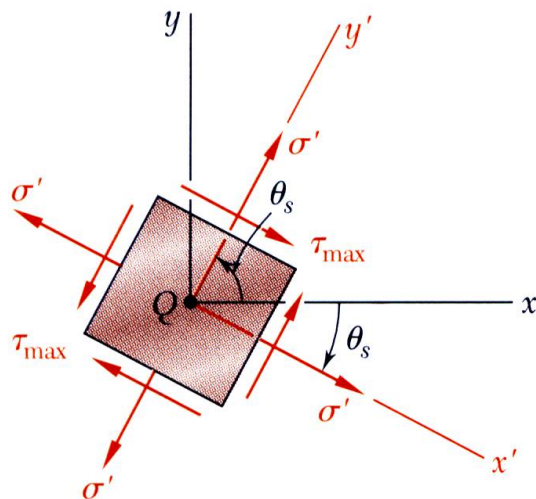
Maximum shearing stress occurs for $\sigma_{x'} = \sigma_{ave}$

$$\tau_{\max} = R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \quad (8)$$

$$\tan 2\theta_s = -\frac{\sigma_x - \sigma_y}{2\tau_{xy}} \text{ (by differentiating eq. 4)} \quad (9)$$

Note: defines two angles separated by 90° and offset from θ_p by 45°

$$\sigma' = \sigma_{ave} = \frac{\sigma_x + \sigma_y}{2}$$



Example 1

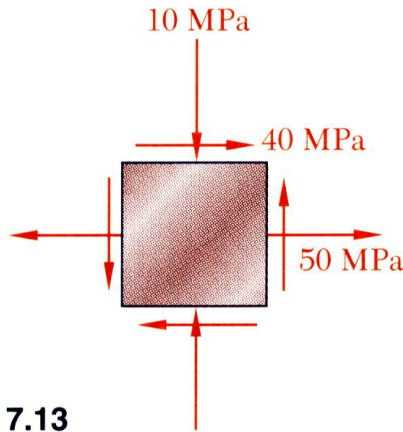


Fig. 7.13

For the state of plane stress shown, determine (a) the principal planes, (b) the principal stresses, (c) the maximum shearing stress and the corresponding normal stress.

SOLUTION:

- Find the element orientation for the principal stresses from

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

- Determine the principal stresses from

$$\sigma_{\max, \min} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

- Calculate the maximum shearing stress with

$$\tau_{\max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma' = \frac{\sigma_x + \sigma_y}{2}$$

Example 1

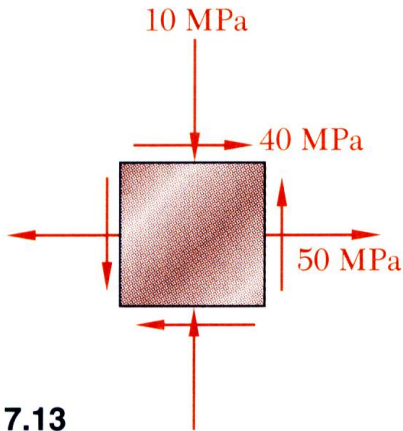


Fig. 7.13

SOLUTION:

- Find the element orientation for the principal stresses from

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} = \frac{2(+40)}{50 - (-10)} = 1.333$$

$$2\theta_p = 53.1^\circ, 233.1^\circ$$

$$\theta_p = 26.6^\circ, 116.6^\circ$$

$$\sigma_x = +50 \text{ MPa} \quad \tau_{xy} = +40 \text{ MPa}$$

$$\sigma_y = -10 \text{ MPa}$$

- Determine the principal stresses from

$$\sigma_{\max, \min} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= 20 \pm \sqrt{(30)^2 + (40)^2}$$

$$\sigma_{\max} = 70 \text{ MPa}$$

$$\sigma_{\min} = -30 \text{ MPa}$$

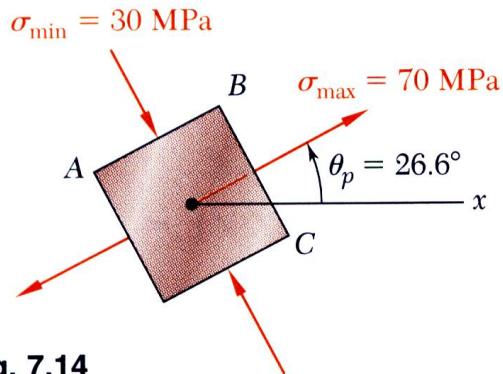


Fig. 7.14

Example 1

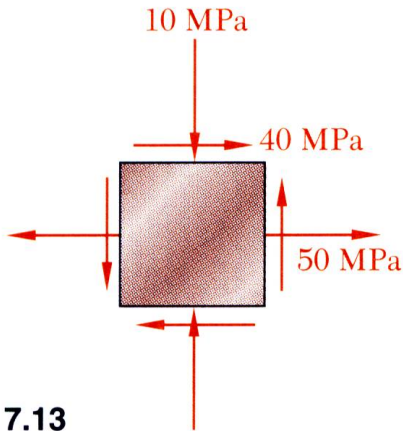


Fig. 7.13

$$\sigma_x = +50 \text{ MPa} \quad \tau_{xy} = +40 \text{ MPa}$$

$$\sigma_y = -10 \text{ MPa}$$

- Calculate the maximum shearing stress with

$$\tau_{\max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \sqrt{(30)^2 + (40)^2}$$

$$\tau_{\max} = 50 \text{ MPa}$$

$$\theta_s = \theta_p - 45^\circ$$

$$\theta_s = -18.4^\circ, 71.6^\circ$$

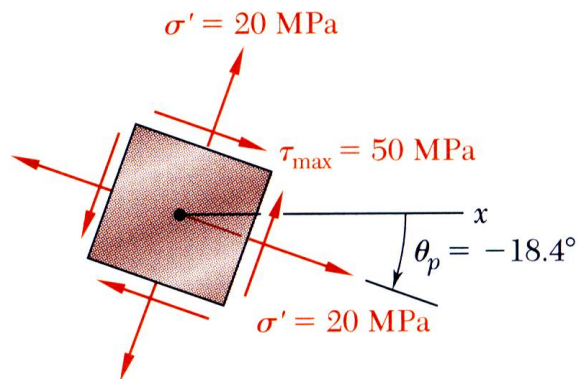


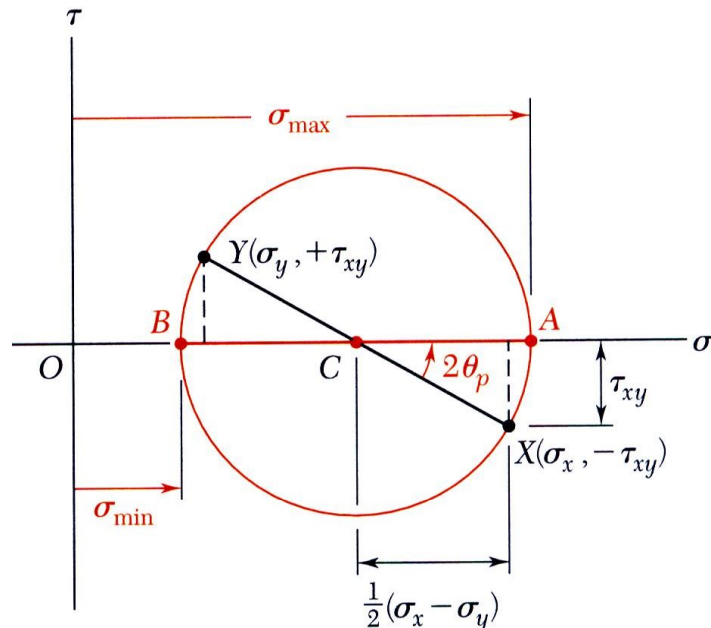
Fig. 7.16

- The corresponding normal stress is

$$\sigma' = \sigma_{ave} = \frac{\sigma_x + \sigma_y}{2} = \frac{50 - 10}{2}$$

$$\sigma' = 20 \text{ MPa}$$

Mohr's Circle for Plane Stress



- With the physical significance of Mohr's circle for plane stress established, it may be applied with simple geometric considerations. Critical values are estimated graphically or calculated.
- For a known state of plane stress $\sigma_x, \sigma_y, \tau_{xy}$ plot the points X and Y and construct the circle centered at C .

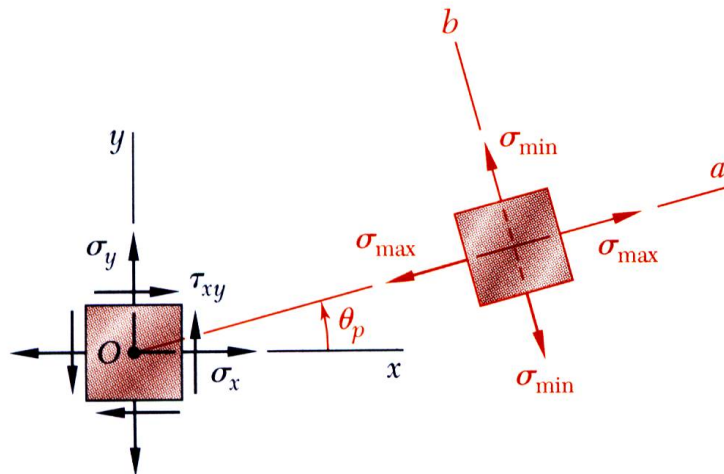
$$\sigma_{ave} = \frac{\sigma_x + \sigma_y}{2} \quad R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

- The principal stresses are obtained at A and B .

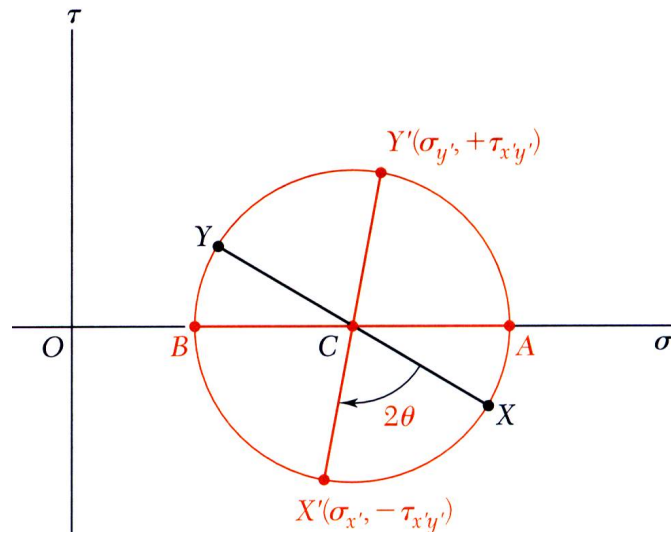
$$\sigma_{\max, \min} = \sigma_{ave} \pm R$$

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

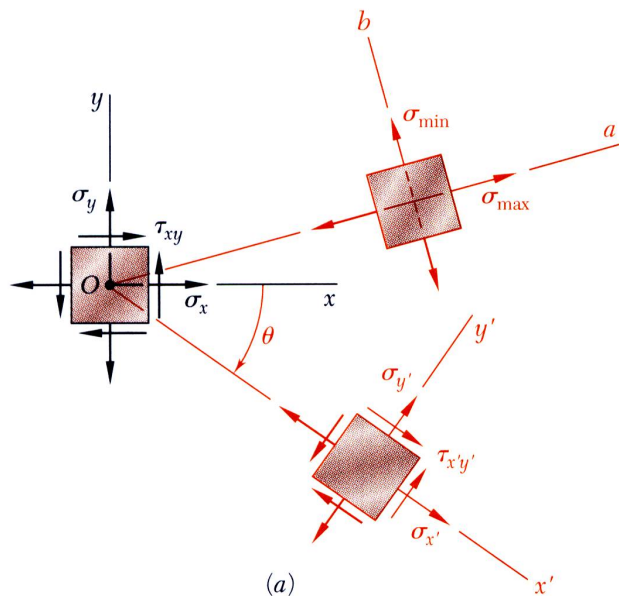
The direction of rotation of Ox to Oa is the same as CX to CA .



Mohr's Circle for Plane Stress



- With Mohr's circle uniquely defined, the state of stress at other axes orientations may be depicted.
- For the state of stress at an angle θ with respect to the xy axes, construct a new diameter $X'Y'$ at an angle 2θ with respect to XY .
- Normal and shear stresses are obtained from the coordinates $X'Y'$.



Example 2

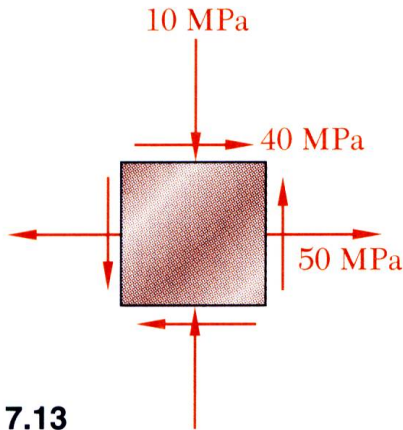
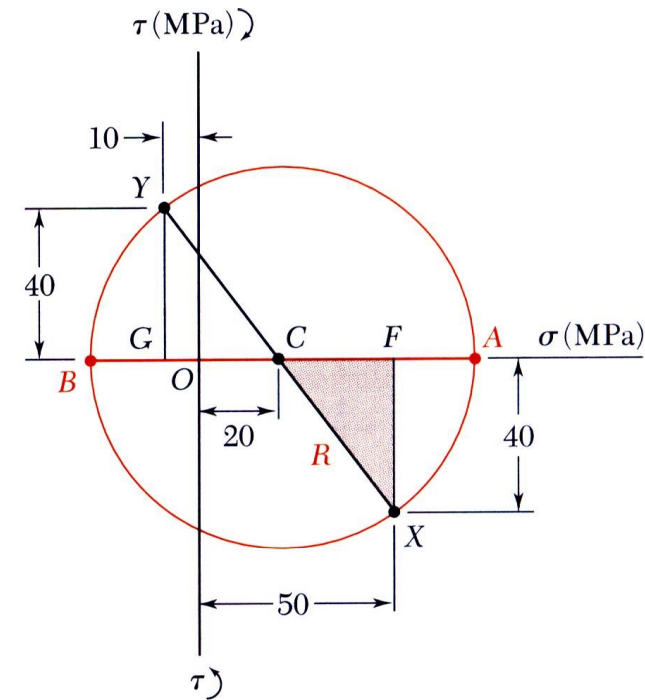


Fig. 7.13

For the state of plane stress shown,
 (a) construct Mohr's circle, determine
 (b) the principal planes, (c) the
 principal stresses, (d) the maximum
 shearing stress and the corresponding
 normal stress.



SOLUTION:

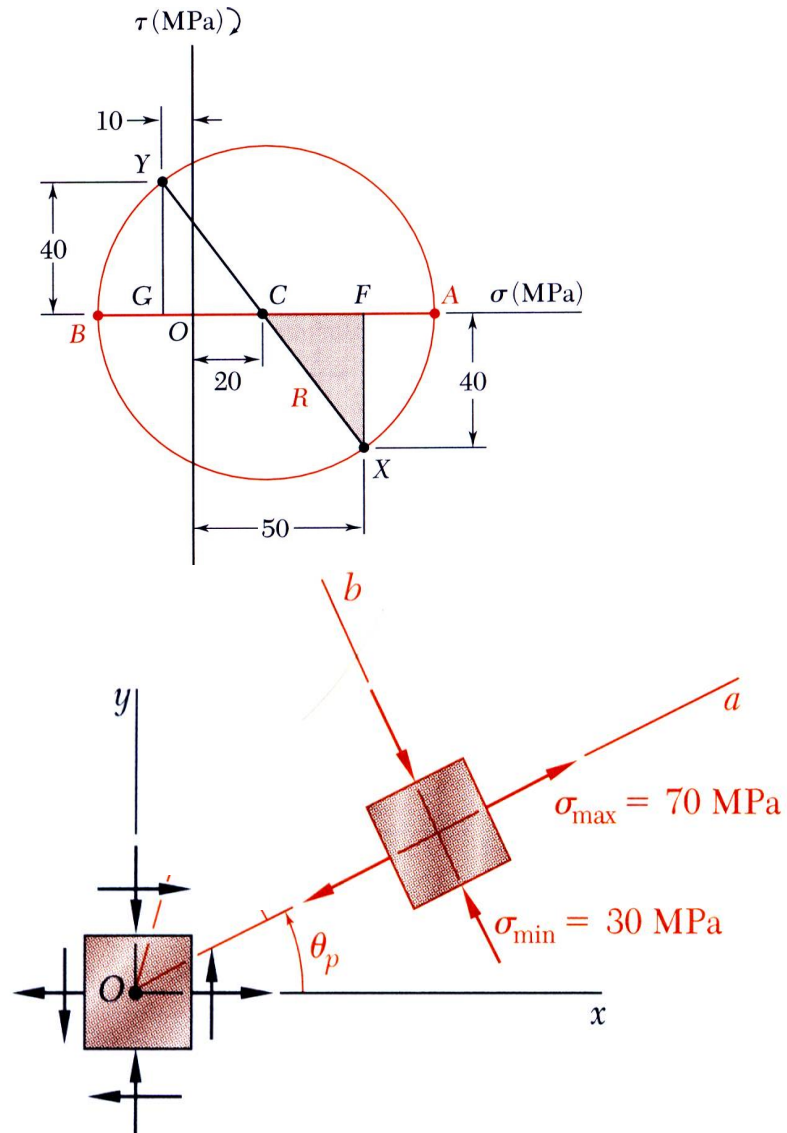
- Construction of Mohr's circle

$$\sigma_{ave} = \frac{\sigma_x + \sigma_y}{2} = \frac{(50) + (-10)}{2} = 20 \text{ MPa}$$

$$CF = 50 - 20 = 30 \text{ MPa} \quad FX = 40 \text{ MPa}$$

$$R = CX = \sqrt{(30)^2 + (40)^2} = 50 \text{ MPa}$$

Example 2



- Principal planes and stresses

$$\sigma_{\max} = OA = OC + CA = 20 + 50$$

$$\sigma_{\max} = 70 \text{ MPa}$$

$$\sigma_{\min} = OB = OC - BC = 20 - 50$$

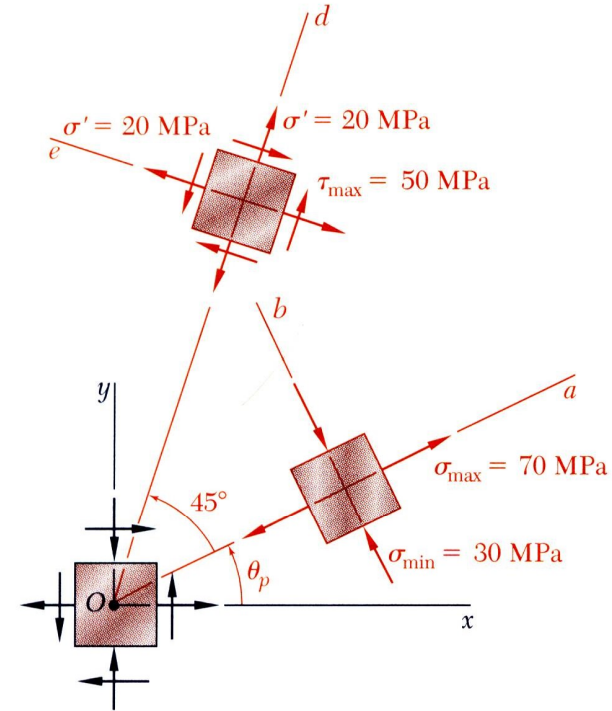
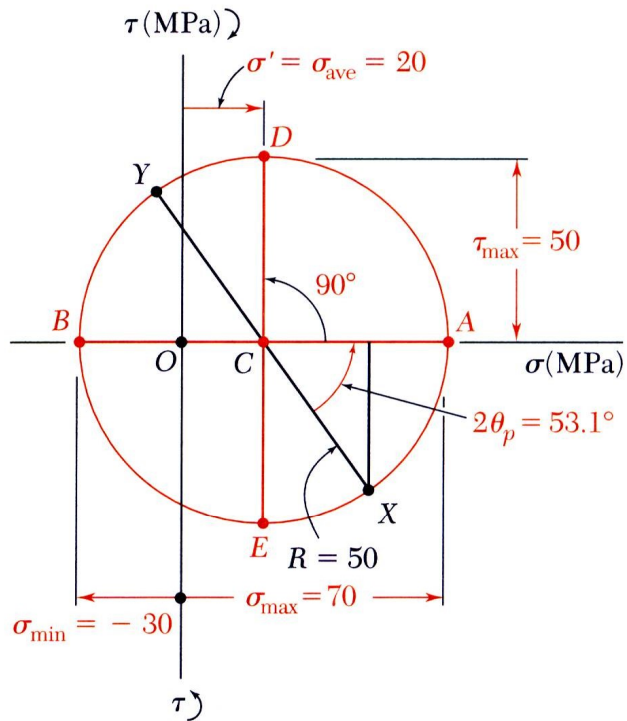
$$\sigma_{\min} = -30 \text{ MPa}$$

$$\tan 2\theta_p = \frac{FX}{CF} = \frac{40}{30}$$

$$2\theta_p = 53.1^\circ$$

$$\theta_p = 26.6^\circ$$

Example 2



- Maximum shear stress

$$\theta_s = \theta_p + 45^\circ$$

$$\theta_s = 71.6^\circ$$

$$\tau_{\max} = R$$

$$\tau_{\max} = 50 \text{ MPa}$$

$$\sigma' = \sigma_{\text{ave}}$$

$$\sigma' = 20 \text{ MPa}$$